Architectural Style Justification

Corporate Mental Wellness & Counseling Platform (Problem Statement #14)

1. Scenario Review

The platform lets employees assess stress, book confidential sessions with licensed therapists and track wellness, without employer identity disclosure. Key functional needs: pseudo-ID generation, assessment scoring, booking, wellness tracking and therapist scheduling. Key non-functional challenges: **security/privacy** (NFR-001), **performance and timezone accuracy** (NFR-002, p95 ≤ 2 s at 2,000 users) and usability for hesitant users.

2. Comparison of Architectural Styles

Style	Strengths for this scenario	Weaknesses for this scenario
Layered	Clear separation of concerns; strict layer boundaries make it easy to enforce that PII never crosses to the business and presentation layers; simple, consistent bookings (single transaction, no double-booking).	Layer overhead; tight coupling between layers; harder to scale one part independently.
Microservices	Independent scaling and fault isolation; the identity vault can be fully isolated.	High operational complexity; network latency; distributed data consistency makes slot locking hard; more attack surface for sensitive data; heavy for the expected scale.
Client-Server	Centralized control and simple deployment; easy data consistency.	Single point of failure; scalability bottleneck; no internal separation, so PII isolation is weak.

3. Selected Architecture: Layered (N-tier) with an Isolated Identity Component

Layered architecture is selected: Presentation, Business Logic and Data layers, with the identity mapping kept as a separate, access-restricted component in the data tier. Reasons: (1) **Privacy by design**: the pseudo-ID is issued in one layer and only the pseudo-ID is passed upward, so therapist and employer views cannot reach PII (FR-001, NFR-001). (2) **Consistency**: booking and slot locking run in one database transaction, preventing double-booking (FR-003) without distributed-transaction complexity. (3) **Fit to scale**: 2,000 concurrent users is well served by a horizontally scaled stateless business layer plus caching, so microservices' cost is not justified. (4) **Maintainability**: simple structure for a small team and easier security audits. The trade-off, layer overhead, is accepted and reduced with caching and connection pooling.

4. Components and Interfaces

Component	Layer	Provided interface	Required interface
Web/Mobile Client	Presentation	-	IAuth, IAssessment, IScheduling (HTTPS/REST)
Identity & Pseudo-ID Service	Business	IAuth (login, token, pseudo-ID)	IDataAccess (identity vault)
Assessment & Wellness Module	Business	IAssessment (score, trends)	IAuth, IDataAccess
Appointment Scheduler	Business	IScheduling (slots, book, cancel; UTC + timezone conversion)	IAuth, IDataAccess, INotify
Encrypted Data Repository	Data	IDataAccess (SQL over TLS)	-
Notification Service	Business	INotify (confirmations, pseudo-ID only)	-

Assembly connectors link each provided interface (ball) to the matching required interface (socket): Client → IAuth/IAssessment/IScheduling, and Identity/Assessment/Scheduler → IDataAccess, Scheduler → INotify.

5. Security and Performance Implications

Security: TLS on all interfaces; AES-256 encryption at rest; the identity vault is reachable only through IAuth; role-based, audit-logged access; employer reports aggregated for groups of 10 or more. Risk: a compromised business layer could reach the vault, mitigated by least-privilege credentials and network segmentation. **Performance:** the business layer is stateless and scaled behind a load balancer; slot searches are cached and invalidated on booking; all times are stored in UTC and converted at the Presentation layer, giving accurate timezone and daylight-saving handling. Risk: the database could become a bottleneck, mitigated with read replicas and indexing. If the platform later outgrows this design, the loosely coupled components can be extracted into services.